Reinforcement Learning-Based Energy-Aware Fault-Tolerant Routing in Underwater Acoustic Networks

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Abstract

Underwater Acoustic Sensor Networks (UASNs) are used for applications such as ocean monitoring, underwater exploration, and environmental monitoring. These networks mainly use sound signals to send data because radio signals do not travel well through water. The sensor nodes are usually battery powered, and replacing or recharging their batteries is difficult after deployment. So we need to use the energy efficiently to increase the network lifetime. Data forwarding uses a lot of energy, especially when the same nodes are used again and again to forward data. This can drain their batteries quickly, cause the nodes to fail, and eventually lead to the collapse of the whole network. So, the routing process should consider the remaining energy of the nodes when choosing which nodes should forward the data.[1].

Nodes with good connectivity or those closer to the destination may be used again and again, which can drain their energy quickly. When such a node fails, the route may break, causing packet loss, retransmissions, extra energy use, and eventually the collapse of the network.Node movement and unreliable underwater connections can cause more route failures. Existing methods such as Vector-Based Forwarding (VBF), Depth-Based Routing (DBR) and others use different ways to select forwarding nodes [2, 3]. Energy-aware methods also consider the remaining energy or communication cost of nodes [4]. However, many existing methods do not properly handle both energy usage and node or link failures. A better routing method is needed to distribute the forwarding work and maintain communication when failures occur.

The proposed method for this project is **RL Based Energy-Aware Fault-Tolerant Routing** for underwater acoustic sensor networks. This method will choose the forwarding nodes based on their residual energy, link reliability, distance or progress toward the destination and also the availability of node. A forwarding score will be obtained from these parameters and used to select the nodes. This work will be shared among different nodes and nodes with very low energy are avoided in order to save energy. Multiple forwarding candidates will be maintained so that an alternative node can be selected when the primary node fails or becomes unreliable. This can reduce packet loss, retransmissions, and unnecessary route reconstruction.

During each forwarding decision, energy consumption will also be considered. If the residual energy of node is too low or its link is unreliable, that node will not be selected even if it has better distance progress. This helps prevent early failure of important relay nodes and distributes energy usage more evenly. In addition, it also reduces the energy usage and delay caused by repeated route discovery.

Objectives:

- To design a routing method that considers the residual energy, distance to the destination, and link quality to reduce energy use and increase network lifetime.
- To handle node and link failures by detecting failures and selecting an alternative suitable forwarding node.
- To evaluate the proposed method using network lifetime, remaining energy, packet delivery ratio, delay, packet loss, throughput, and routing overhead.

Testing of the proposed method will be done through the simulation at different energy levels, network sizes, node densities, and failure conditions. Performance will be compared with the existing routing methods. The evaluation will focus on reducing energy imbalance, maintaining communication after node failures, increasing packet delivery, and improving network lifetime without excessive routing overhead.

References

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